

Module 66.1: Derivatives Markets

SCHWESERNOTES - BOOK 3

LOS 66.a: define a derivative and describe basic features of a derivative instrument.

A **derivative** is a security that *derives* its value from the value of another security or a variable (such as an interest rate or stock index value) at some specific future date. The security or variable that determines the value of a derivative security is referred to as the **underlying** for the derivative. The value of a derivative at a point in time is derived from the value of the underlying (asset or variable) on which the derivative contract is based.

A relatively simple example of a derivative is a **forward contract** that specifies the price at which one party agrees to buy or sell an underlying security at a specified future date. Consider a forward contract to buy 100 shares of Acme at \$30 per share three months from now:

- Acme shares are the **underlying asset** for the forward contract.
- \$30 is the **forward price** in the contract.
- The date of the future transaction, when the shares will be exchanged for cash, is referred to as the **settlement date** (maturity date) of the forward contract.
- 100 shares is the **contract size** of the forward contract.
- The forward price is set so that the forward contract has zero value to both parties at contract initiation; neither party pays at the initiation of the contract.

We can examine three outcomes for the price of Acme shares at settlement:

1. The **spot price**, or the market price of the underlying, is \$30, equal to the forward price of \$30.

Neither party has profits or losses on the forward contract. Ignoring transactions costs, the party selling Acme shares could buy them back at \$30 per share and the party buying the shares could sell them at \$30 per share.

2. The spot price of Acme shares is \$40, greater than the forward price of \$30.

The party buying 100 Acme shares for \$3,000 at settlement can sell those shares at the spot price for \$4,000, realizing a profit of \$1,000 on the forward contract. The party that must deliver the Acme shares delivers shares with a market value of \$4,000 and receives \$3,000, realizing a \$1,000 loss on the forward contract.

3. The spot price of Acme shares is \$25 at settlement, less than the forward price of \$30.

The party buying 100 Acme shares for \$3,000 at settlement can sell those shares at the market price of \$25 to get \$2,500 and realize a loss of \$500 on the forward contract. The party that must deliver the Acme shares delivers shares with a market value of \$2,500 and receives \$3,000, realizing a \$500 gain on the forward contract.

To summarize, the buyer of the shares in a forward contract will have gains when the market price of the shares at settlement is greater than the forward price, and losses when the market price of the shares at settlement is less than the forward price. The party that must deliver the shares in a forward contract will have gains when the market price of the shares at settlement is less than the forward price, and losses when the market price of the shares at settlement is greater than the forward price. The gains of one party equal the losses of the other party at settlement.

We refer to the party that agrees to buy the underlying asset in a forward contract as the buyer of the forward. The buyer of the forward gains when the price of the underlying increases (and loses when it falls), similarly to a long position in the underlying. In this case we say the forward buyer has long exposure to the underlying, while the seller of the underlying has short exposure to the underlying, gaining when the price of the underlying decreases and losing when the value of the underlying increases.

In practice, a forward contract may be a **deliverable contract**. In our forward contract example, this means that the payment and the shares must be exchanged at the settlement date. A **cash-settled contract** specifies that only the gains and losses from the forward contract are exchanged at settlement. In our example above, with a share price of \$25, cash settlement would require the buyer of Acme shares to pay \$500 to the seller of Acme shares at settlement.

Ignoring the transaction cost of buying and selling shares in the market, the gains and losses to the parties in our forward contract example are economically equivalent under the two alternative settlement methods.

We can view a derivatives contract as a way to transfer risk from one party to another. Consider a situation where the share seller in the forward contract owns 100 shares of Acme. She has existing risk because the future price of Acme shares is uncertain; the share price three months from now is a random variable. If she enters the forward contract from our example, she will receive \$3,000 for her shares at settlement, regardless of their market price. This effectively transfers her existing Acme price risk to the buyer of the shares in the forward contract.

When a party to a derivative contract has an existing risk that is transferred to another party, we say that party has **hedged** (offset, reduced) their existing risk. If the risk of the forward contract exactly matches an existing risk, then the forward contract can be used to fully hedge the existing risk. If a derivative is used to reduce, but not entirely offset, an existing risk, we say the existing risk is **partially hedged**.

If the Acme share buyer in our example has no existing Acme price risk, she clearly increases her risk by entering into the forward contract. In this situation, the Acme share buyer is said to be **speculating** on the future price of Acme shares.

You may have realized our share seller could have achieved her goal of eliminating her Acme price risk by simply selling her shares (a **cash market transaction**). Derivatives have potential advantages over cash market transactions:

- Investors can gain exposure to a risk at low cost, effectively creating a highly leveraged investment in the underlying.
- Transaction costs for a derivatives position may be significantly lower than for the equivalent cash market trade.
- Initiating a derivatives position may have less impact on market prices of the underlying, relative to initiating an equivalent position in the underlying through a cash market transaction.

Underlying Assets and Variables

The underlying for a derivative is most often a stock or bond price, the level of a stock or bond index, or an interest rate. Here, we give examples of different underlying assets and variables for derivative contracts and the nature of the risks they transfer or modify. We present more details about derivatives based on these underlying assets in subsequent readings.

- A bond, for example a forward contract on a 30-year U.S. Treasury bond or other specific bond. The risk involved is the uncertainty about future bond prices.
- An index, for example the S&P 500 Index or the Citi Goldman Sachs Investment Grade Corporate Bond Index. The risk involved is the uncertainty about the future value of the index at a specific date. A portfolio manager can reduce the risk of a portfolio of large U.S. stocks for a period of time by selling a forward on the S&P 500 index. An investor can gain long exposure to a portfolio of high-grade corporate bonds, quickly and at low cost, by buying a forward on the index.
- A currency, for example British pounds (GBP). A U.S. manufacturer that expects a large payment in GBP in six months can offset the uncertainty about the USD value of this payment by selling a forward contract on the expected amount of GBP. A UK manufacturer that must make a large payment in USD in one year can offset the uncertainty about the GBP cost of the payment by buying a forward contract on the USD that is priced in GBP.
- An interest rate, for example the 1-year Treasury bill rate. Similar to derivatives based on bonds, except that a higher interest rate means gains for the buyer of an interest rate forward, whereas higher interest rates mean lower bond prices and losses for the buyer of a bond forward.
- Commodities, which are physical assets including **hard commodities** (typically mined or extracted, such as gold and oil) and **soft commodities** (typically grown, such as cotton, coffee, pork, and cattle). A farmer expecting a cotton crop in four months can reduce her cotton price risk by selling cotton forward. A utility that will require thousands of gallons of oil over the next year can reduce its oil price risk by buying oil forwards that settle at various times over the coming year

Professor's Note

It may help you to remember this common rule for hedging risk with futures, which are similar to forwards: "Do in the futures market what you must do in the future." A baking company that must buy wheat in the future should buy wheat futures (or forwards) to reduce the effects of the uncertainty about future wheat prices on their profits. A farmer who must sell wheat at harvest time should sell wheat futures (or forwards) to reduce price risk.

- Credit derivatives include credit default swaps (CDS), in which one party makes fixed periodic payments to another party, which will make a payment only if the underlying credit instrument (or portfolio of such credit securities) suffers a loss in value due to a default by the issuer (borrower of funds) of the credit instrument.
- Derivative contracts are also created with the weather (for farmers, energy producers, travel and tourism companies), cryptocurrencies, or longevity (for life insurers or annuity providers) as the underlying asset.

Along with forwards and futures, derivatives types that we will cover in the remainder of our derivatives readings are:

- *Options*: Put options give the buyer the right (but not the obligation) to sell the underlying for a specific price in the future. Call options give the buyer the right (but not the obligation) to buy the underlying for a specific price in the future.
- *Swaps*: In a simple interest rate swap, one party agrees to make periodic payments on a given amount at a fixed interest rate, and the other agrees to make periodic interest payments on the same given amount, but at an interest rate based on a future market reference rate (MRR). The resulting cash flows are equivalent to one party (the fixed-rate payer) borrowing at a fixed rate and using the proceeds to buy a floating-rate bond.

LOS 66.b: describe the basic features of derivative markets, and contrast over-the-counter and exchange-traded derivative markets.

Exchange-Traded Derivatives

Centralized physical exchanges provide markets for futures contracts, some options contracts, and some other derivative contracts. The largest by volume of trades are the National Securities Exchange (India), the B3 market (Brazil), and the CME Group (the United States).

Exchange-traded derivatives are standardized and backed by a central clearinghouse. The exchange specifies the terms of each of the derivative contracts that will be traded and rules for trading on the exchange.

A **central clearinghouse (CCH)** essentially takes the opposite position to each side of a trade (called **novation**), guaranteeing the payments promised under the contract. The CCH requires deposits from both participants when a trade is initiated, and additional deposits for accounts that decline in value, to support its guarantee and minimize counterparty credit risk.

Exchange members (dealers or market makers) buy and sell derivatives at slightly different prices and primarily earn trading profits from the bid/ask spreads between buy and sell prices, rather than from holding (speculating on) specific derivatives positions, although they may hold such positions from time to time to meet customer needs.

The standardization of contracts allows exchange-traded derivatives to be more liquid and more transparent to market participants, compared to customized derivatives. A market participant who has taken a position in exchange-traded derivatives can easily exit that position by entering into a contract with a position opposite to their existing derivatives. Standardization of contracts reduces trading costs compared to customized derivative contracts.

Standardization also facilitates the clearing and settlement of trades. Clearing refers to executing the trade, recording the participants, and handling the exchange of any required payments. Settlement refers to the exchange of underlying assets or payments of the final amounts due at contract settlement (maturity).

Dealer (OTC) Markets for Derivatives

Forwards, most swaps, and some options are custom instruments created and traded by dealers in a market with no central location. Some dealer markets are quite structured (e.g., the Nasdaq market), while others are not. A dealer market with no central location is referred to as an over-the-counter (OTC) market. OTC markets are largely unregulated and less transparent than exchange markets. In OTC markets with no central clearinghouse, each side of a trade faces counterparty credit risk. Dealers (market makers) make derivatives trades with end users of derivatives and may also trade with each other to reduce their exposures to changes in prices of underlying assets.

OTC derivatives contracts can be customized to fit the needs of an end user regarding contract size, definition of the underlying, settlement date, whether the contract is deliverable or cash settled, and other relevant details. Users trying to gain or hedge a specific risk use OTC derivatives when a standardized derivative contract will not meet their needs (including a desire for privacy).

After the financial crisis of 2008, regulators worldwide instituted a **central clearing mandate** requiring that, for many swap trades, a **central counterparty (CCP)** takes on the counterparty credit risk of both sides of a trade, similar to the role of a central clearinghouse. As an example, multiple dealers record their swap trades on a **swap execution facility (SEF)**. When a dealer makes a swap trade, that information is sent to the SEF and the CCP replaces the trade with two trades, with the CCP as the counterparty to both of them, reducing counterparty risk. The downside of this structure is that counterparty risks are concentrated rather than distributed among financial intermediaries.

The following offers a summary of the primary differences between exchange-traded and OTC derivatives.

Exchange-traded derivatives are:

- Traded at a centralized location, an exchange.
- Traded by exchange members (market makers).
- Based on standardized contracts and have lower trading costs.
- Subject to the trading rules of the exchange (i.e., are more regulated).
- Backed by the central clearinghouse to minimize counterparty credit risk. They also require deposits by both parties at initiation, and additional deposits when a position decreases in value.
- More liquid.
- More transparent, as all transactions are known to the exchange and to regulators.

OTC derivatives (not subject to the central clearing mandate) are:

- Custom instruments.
- Less liquid and have higher transaction costs.
- Less transparent.
- Subject to counterparty risk.
- More difficult to clear and settle.
- Subject to higher trading costs.
- Not subject to requirements for the deposit of collateral.

Derivatives in dealer markets that are subject to the central clearing mandate have reduced counterparty risk, are subject to more disclosure of trades, and are easier to clear and settle, but are still customizable and are contracts with dealers or financial intermediaries.

Reading 66: Key Concepts

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LOS 66.a

A derivative is a security that derives its value from value of another security or variable at a specific future date. The security or variable that determines the value of a derivative security is referred to as the underlying.

Basic features of a derivative include the underlying, the price specified in the contract, the contract size, and the settlement date. The price is typically set so the contract has zero value at initiation to both parties. Contracts may be deliverable or cash-settled.

LOS 66.b

Exchange-traded derivatives are standardized and backed by a central clearinghouse that takes the opposite position to each side of a trade, guaranteeing the payments promised under the contract.

Over-the-counter (OTC) derivatives can be customized to fit the needs of the counterparties. OTC markets are largely unregulated and less transparent than exchange markets. Some OTC markets are subject to a central clearing mandate that reduces counterparty credit risk.